

## **Unnamed\_Unit**

### **Unit 4 - Variables**

#### **PDF Documents**

##### **PDF Document 1: 1738153978105-Chapter 10 - Unit 4 - Variables.pdf**

This PDF document is included in the unit materials.

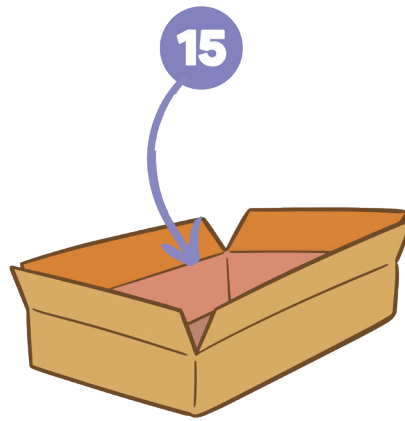
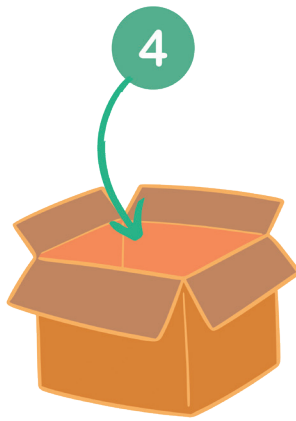


## UNIT 4 VARIABLES

### Variables

Variables are like labeled boxes in a computer where we keep information we need, like numbers or names. They help us remember and change things in our code, making it smarter.

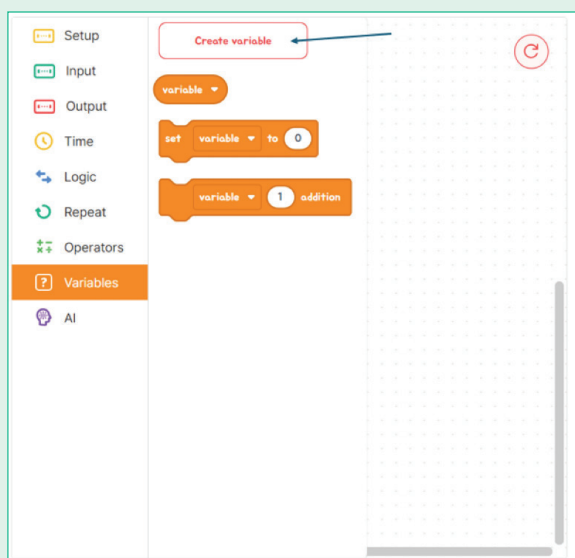
## ✓ VARIABLES

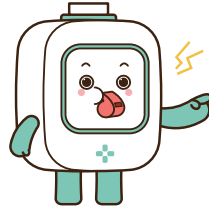


LET'S MAKE A VARIABLE IN CODE EDITOR:

### Step1

Go to the Variables section and click "Create Variable."





### Step3

After typing the name, click "Make" to create it.

### Step2

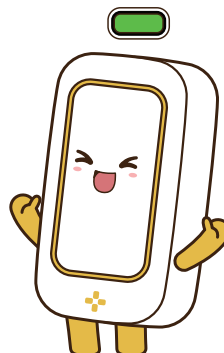
Type the name for your variable (like "mode").

### Step4

You'll then see your new variable in the list.

### Step5

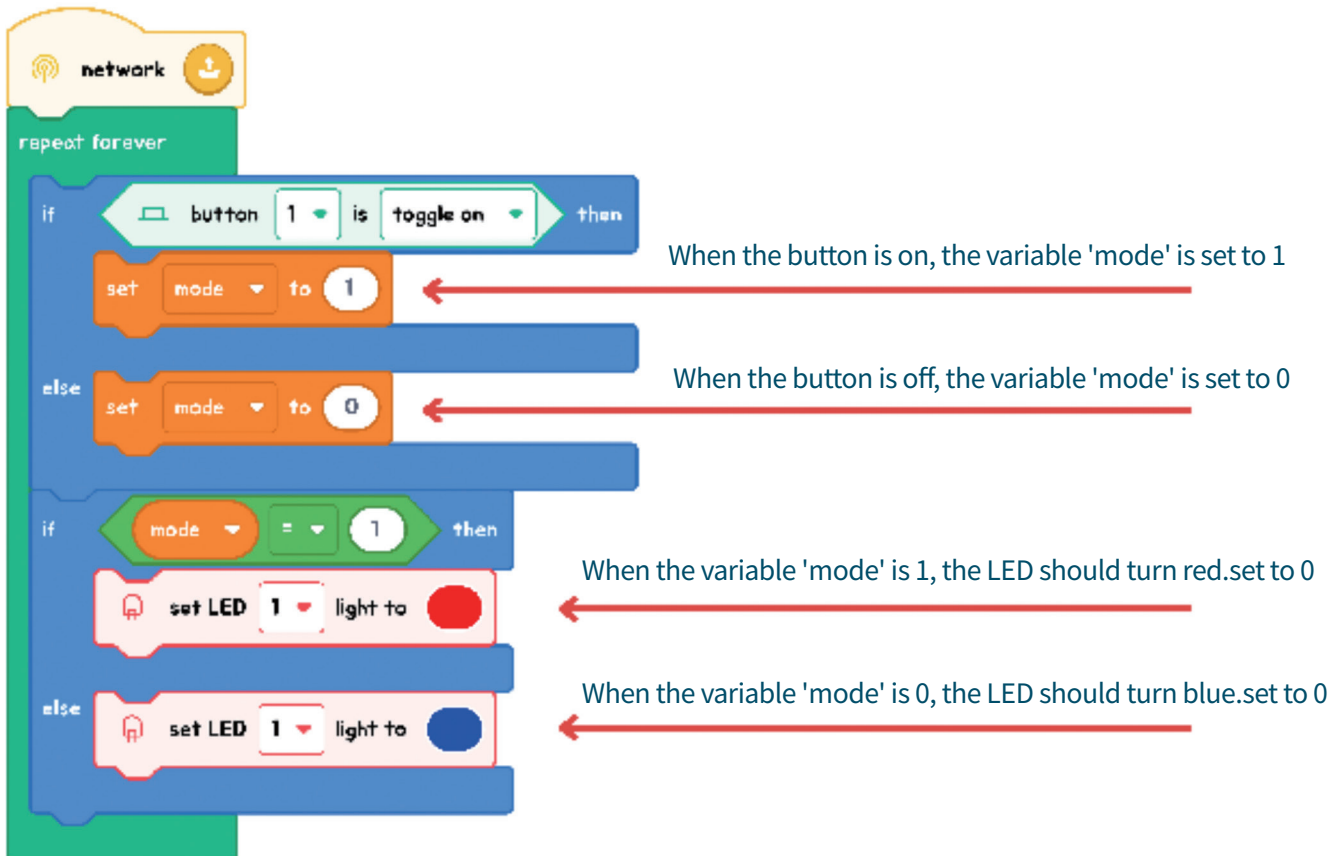
Click on the area where variables are shown to see the one you made.





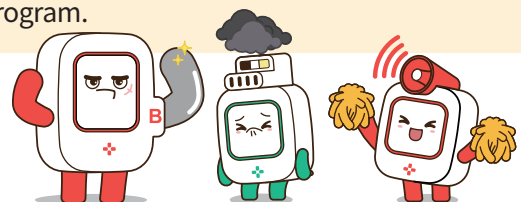
## EXPLANATION WITH CODE EXAMPLE

In our code, we're using a button and an LED light along with a variable named 'mode'. Here's what happens in our program:



### Code Insight

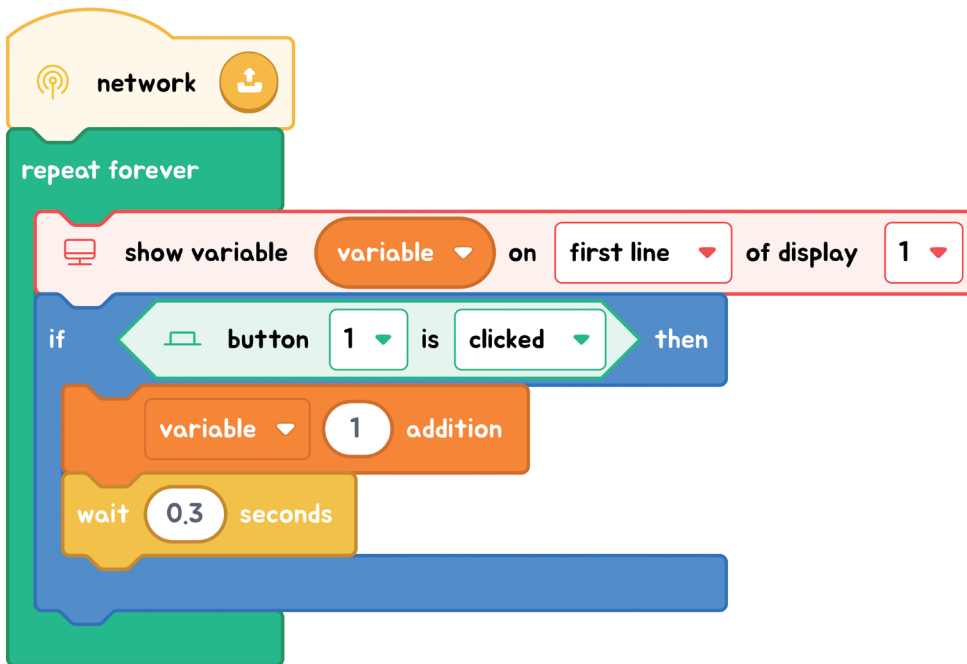
- 1. Variable 'mode':** Think of 'mode' as the program's memory of whether the button is on or off. 'Mode' can be either 1 (on) or 0 (off).
- 2. Checking the Button:** The first part of our code checks the button's state. If the button is on, 'mode' gets set to 1; if it's off, 'mode' goes to 0. This is how the program 'remembers' the button's state.
- 3. Action Based on 'mode':**
  - If 'mode' is 1: The program sees 'mode' is 1 and knows the button was on, so it turns the LED red.
  - If 'mode' is 0: The program finds 'mode' is 0, it makes the LED blue.
- 4. The Role of 'mode':** The variable 'mode' acts as a switch for the LED's color, changing based on the button's state, showing how variables can control actions in a program.



## Exercise

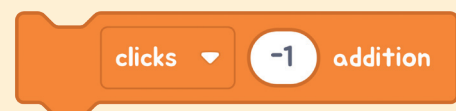
### Exercise: Click Counter challenge

Every time you click the button, we want to add one more to our click counter (that's our variable). Using a 'plus' block in block coding, we can keep track of how many times the button has been clicked. When we first make a variable, it starts at 0, unless we tell it to start at a different number.



#### Code Insight

- Starting Point:** When we begin, our display screen is set to show a number(variable) called "clicks." Think of "clicks" like your score in a video game, which everyone can see on the screen.
- Click to Add:** Each time the button is pressed, we add 1 to our "clicks." Because the display is always showing "clicks," everyone can see the score go up each time the button is clicked.
- Updating the Score:** Just like how a scoreboard updates every time a player scores in a game, our display updates the "clicks" number every time the button is clicked, showing the new, higher score.
- To decrease the number with each button click:** switch to using a subtraction operation in the code.



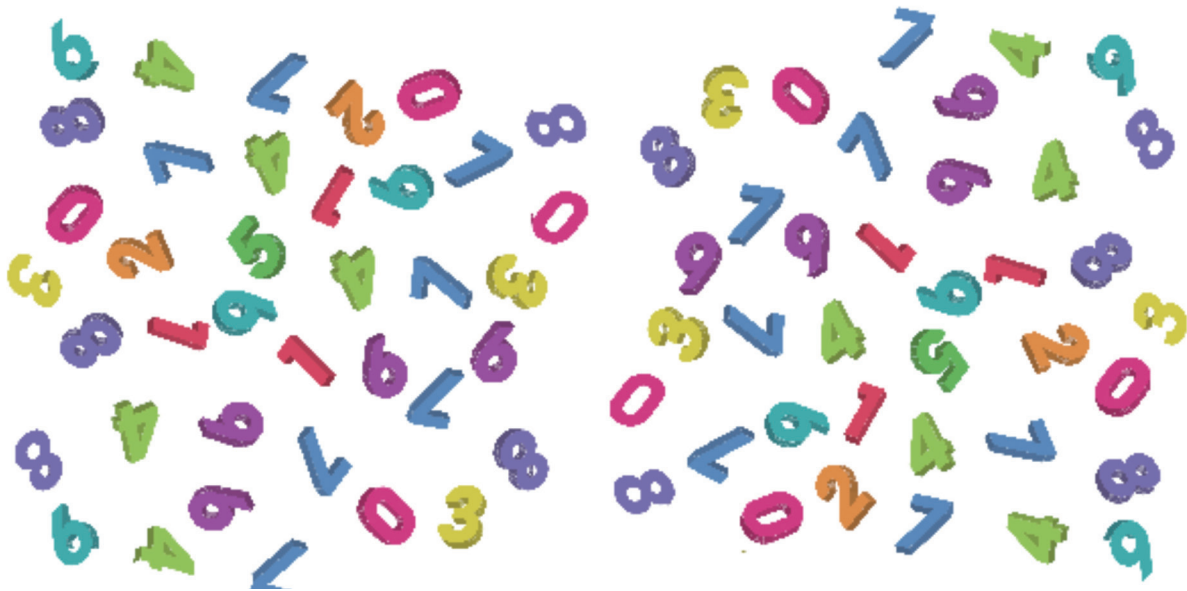


## Random

### WHAT IS 'RANDOM'?

In programming, when we talk about something being "random," we mean it's like flipping a coin, rolling a dice, or drawing a card from a shuffled deck. You don't know what you're going to get each time. It's the computer's way of making a surprise choice.

# ✓ RANDOM



### HOW DOES RANDOM WORK?

Computers are very good at following instructions, but they don't actually make random choices the way humans can. Instead, they use special commands that create "pseudo-random" numbers. These are sequences that seem random enough for most uses even though they're actually coming from a set pattern.

### WHY USE RANDOM?

#### 1. Games

Random numbers can control things like what kind of monster pops out or what treasure you find in a video game, making the game fun and different each time.



## 2. Simulations

When scientists want to see how things might happen differently, like predicting weather or how traffic moves, they use random numbers to represent all the unknowns.

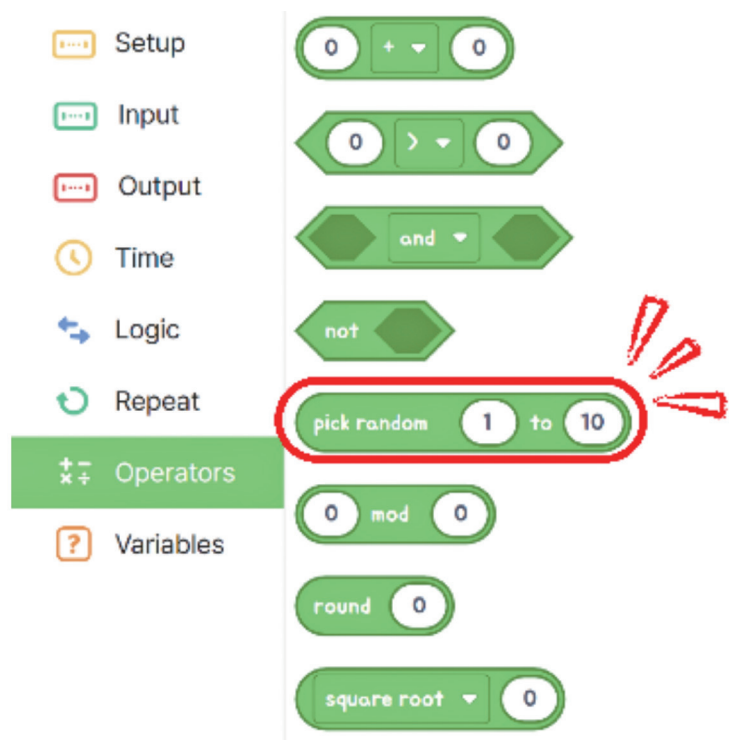


## 3. Safety and Security

When creating passwords or encryption keys to keep data safe, using random numbers helps make sure they're really hard for someone else to guess.

### RANDOM BLOCK IN CODE EDITOR

In the operators section, there is a block for generating random numbers that allows you to specify the desired range.





## Exercise

### Multiple-Choice Questions



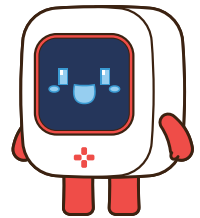
#### QUESTION 1

When comparing "and" with nested if statements, what similarity do they share?

- ☐ A Both require each condition within them to be checked sequentially.
- ☐ B Both require all conditions to be true for the code block to execute.
- ☐ C Both can be replaced by other logical operators in all scenarios.
- ☐ D Both are primarily used to compare numeric values.

#### THINK ABOUT IT!

Do you remember?  
The nested-if statement  
works the same as "AND"!



#### QUESTION 2

How does the "or" operator differ from the "and" operator in programming logic?

- ☐ A The "or" operator requires only one condition to be true for the code block to execute, while the "and" operator requires all conditions to be true.
- ☐ B The "or" operator is used primarily for exclusivity checks, while the "and" is for inclusivity.
- ☐ C The "or" operator evaluates all conditions at once, while the "and" evaluates them sequentially.
- ☐ D The "or" operator can be used interchangeably with the "and" operator without changing the logic of the code.

#### THINK ABOUT IT!

What is the main difference  
between the "AND" and  
"OR"?

